

ECE 100 – Linear Electronic Systems

Lab 1 Report – Passive RLC Circuit Analysis With MATLAB

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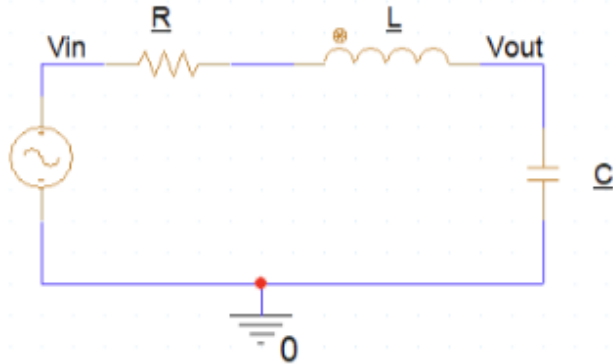
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Abstract

The purpose of this lab is to analyze RLC circuits using MATLAB. By plugging various zeta values (known as the damping ratios) and varying frequencies into the transfer functions, we saw how these frequencies were filtered out or passed through. We concluded that MATLAB is an excellent tool in analyzing signals such as transfer functions for filters.

Experimental Procedures and Results

1. Low-pass RLC circuit:



Problem 1: Low-pass RLC circuit

With an RLC circuit with the voltage taken across the capacitor, we create a low-pass filter. This causes the transfer function to filter out low frequencies. The following below are our calculations by hand and plotting using MATLAB.

Solving the transfer function of the LPF by voltage divider

we get:

$$V_o = \frac{Z_c}{Z_c + Z_L + Z_R} V_i \Rightarrow \frac{1/sC}{1/sC + sL + R} \cdot sC$$
$$H_{LP}(s) = \frac{1}{1 + sRC + s^2LC}$$

Converting into the frequency response, with the values of the natural freq. ω_0 and the damping coefficient ζ , we get

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad \zeta = \frac{R}{2} \sqrt{\frac{C}{L}}$$

$$H(j\omega) = \frac{1}{1 + 2\zeta \frac{j\omega}{\omega_0} + (\frac{j\omega}{\omega_0})^2}$$

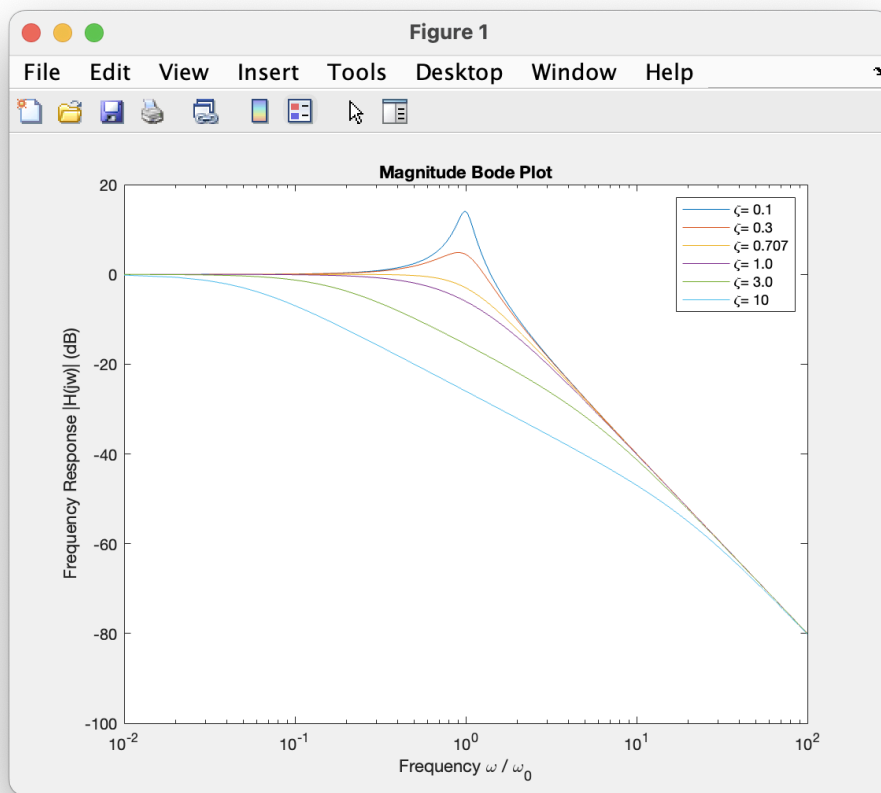


Figure 1: LPF Magnitude Bode Plot

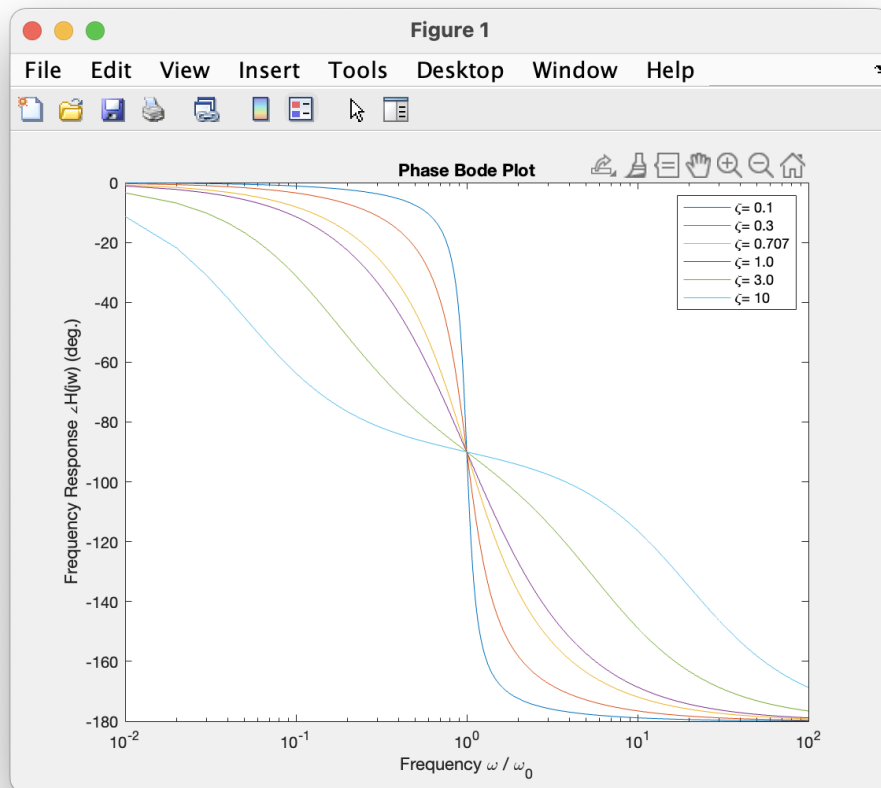


Figure 2 : LPF Phase Bode Plot

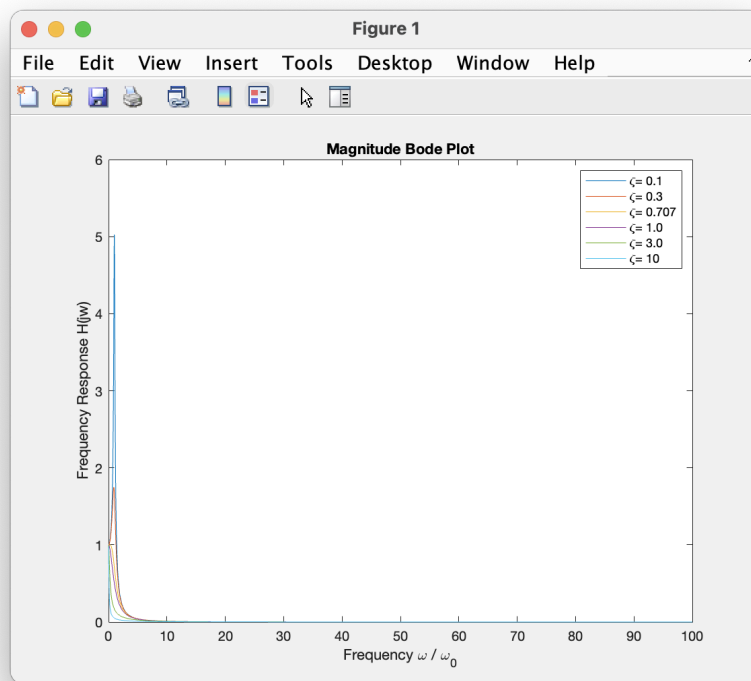


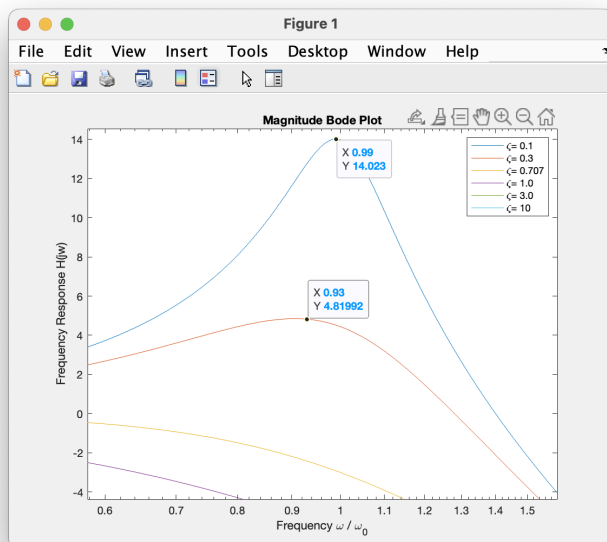
Figure 3 : LPF Linear Bode Plot

$$\omega = \sqrt{1-2\zeta^2}$$

$$|H(j\omega)| = \frac{1}{\sqrt{(1-\omega^2)^2 + (2\zeta\omega)^2}} \Rightarrow \frac{1}{\sqrt{(1-1+2\zeta^2)^2 + 4\zeta^2(1-2\zeta^2)}} \Rightarrow \frac{1}{4\zeta^4 + 4\zeta^2 - 8\zeta^4} = \frac{1}{\sqrt{-4\zeta^4 + 4\zeta^2}}$$

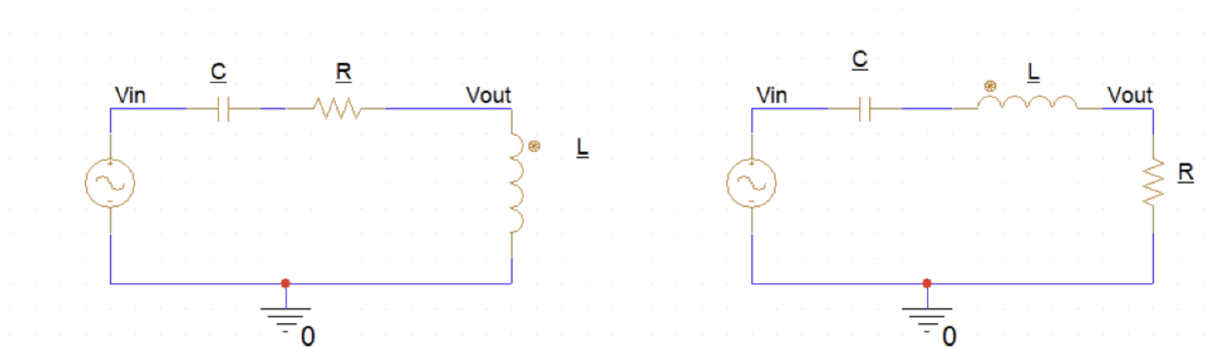
$\zeta = 0.1$
 $|H(j\omega)| = \frac{1}{\sqrt{-4(0.1)^4 + 4(0.1)^2}} = 5.03$
 $20 \log_{10} |H(j\omega)| = 20 \log_{10}(5.05) = 14.03 \text{ dB}$

$\zeta = 0.3$
 $|H(j\omega)| = \frac{1}{\sqrt{-4(0.3)^4 + 4(0.3)^2}} = 1.74$
 $20 \log_{10} |H(j\omega)| = 20 \log_{10}(1.74) = 4.82 \text{ dB}$



The values match the ones from our calculations when finding the peak.

2. High-pass and Band-pass RLC circuits:



Problem 2: High-pass and Band-Pass RLC Circuits

With the High-pass and Band-Pass filters, they take the voltage output across different circuit elements, respectively an inductor and resistor. This causes the transfer function and overall filter to change qualities.

The frequency responses for the HPF and BPF are:

$$H_{HP}(j\omega) = \frac{\left(\frac{j\omega}{\omega_0}\right)^2}{1 + 2\zeta\left(\frac{j\omega}{\omega_0}\right) + \left(\frac{j\omega}{\omega_0}\right)^2}$$

$$H_{BP}(j\omega) = \frac{2\zeta\left(\frac{j\omega}{\omega_0}\right)}{1 + 2\zeta\left(\frac{j\omega}{\omega_0}\right) + \left(\frac{j\omega}{\omega_0}\right)^2}$$

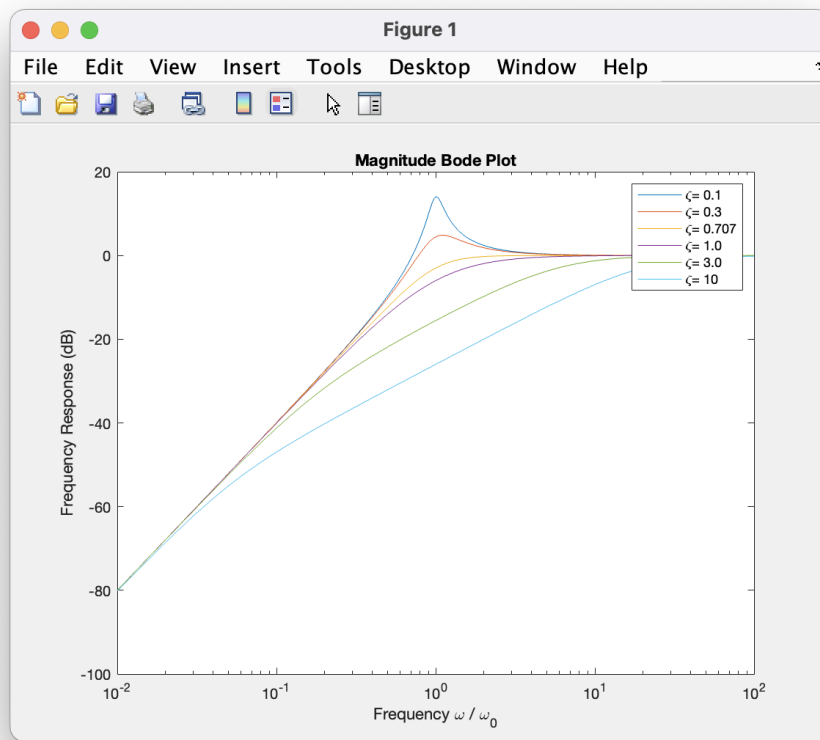


Figure 4 : HPF Magnitude Bode Plot

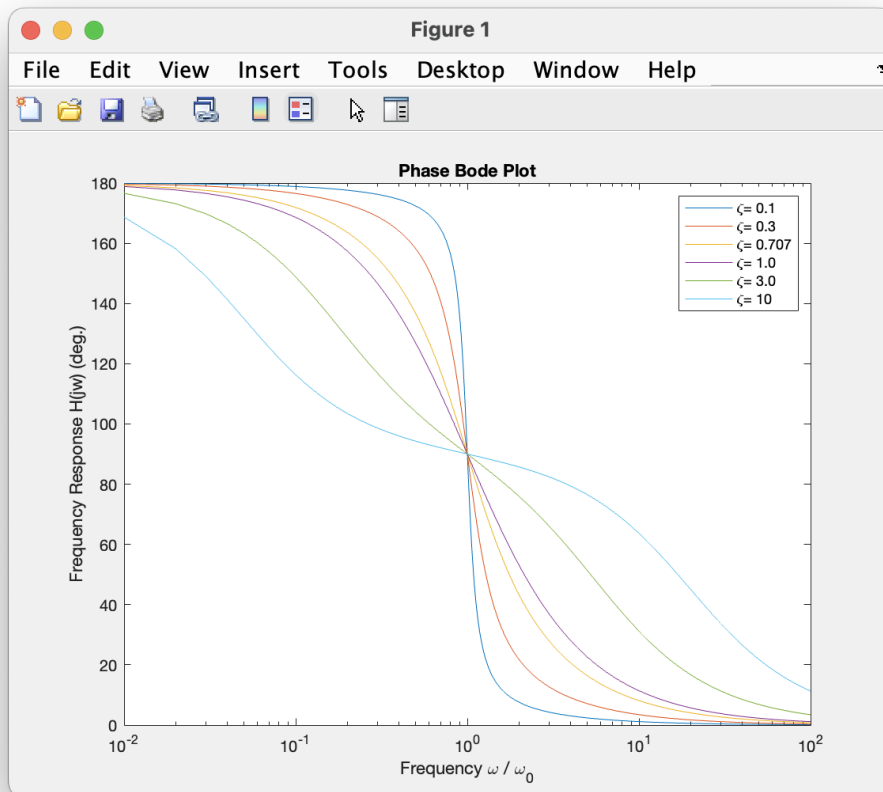


Figure 5 : HPF Phase Bode Plot

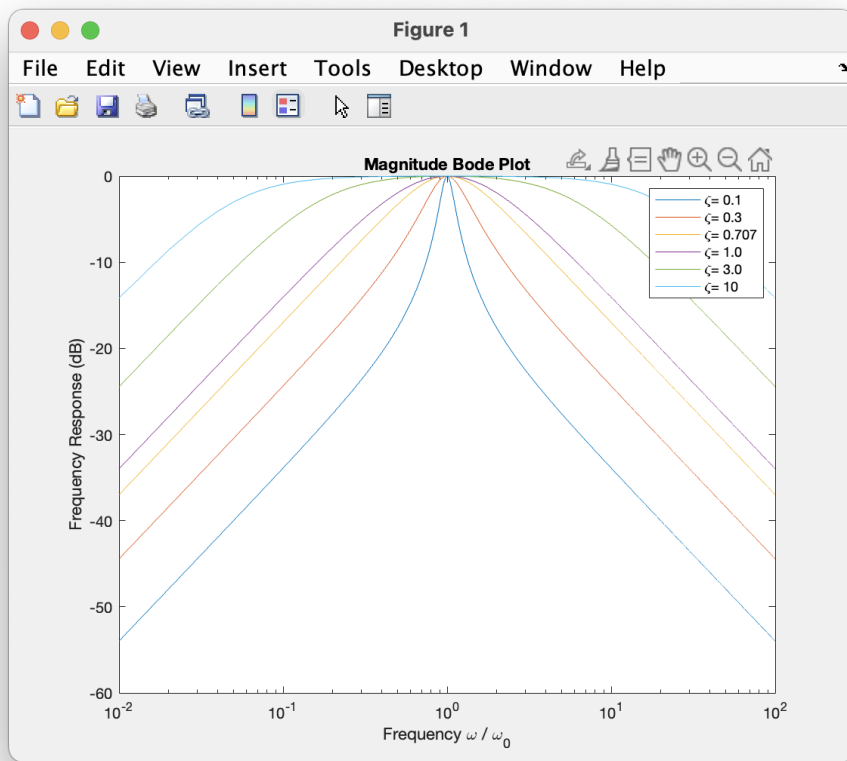


Figure 6: BPF Magnitude Bode Plot

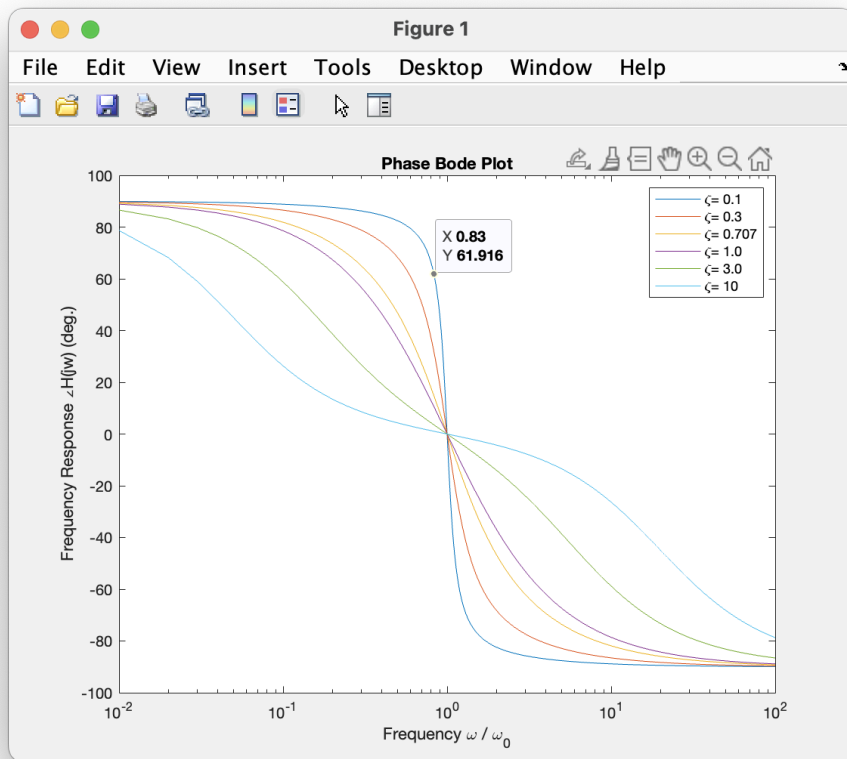
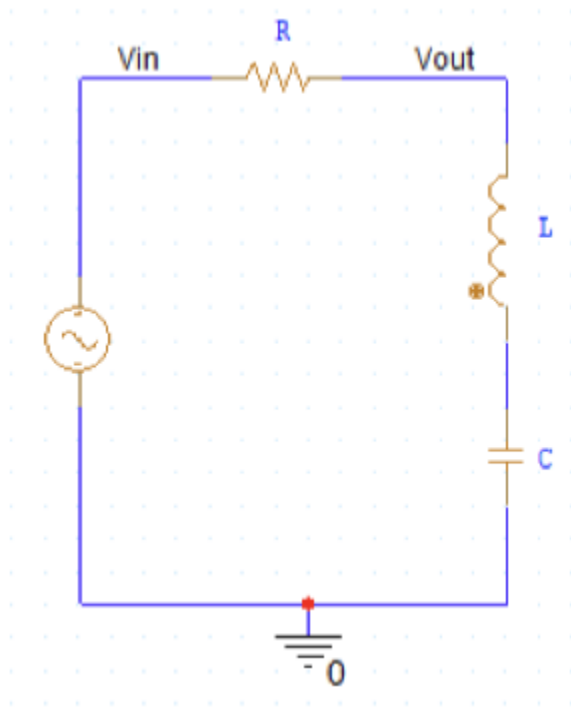


Figure 7: BPF Phase Bode Plot

Problem 3: Mystery Filter



For this mystery filter we would solve it the same way as before, with the only difference being the numerator on top. The transfer function then equals to:

$$H_{MF}(s) = \frac{s^2 LC + 1}{1 + sRC + s^2 LC}$$

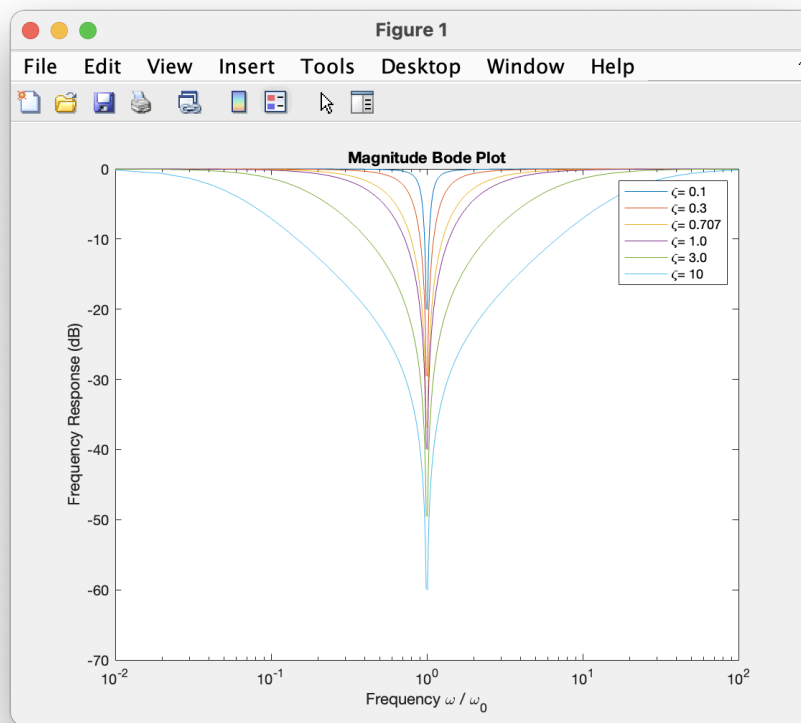


Figure 8: MF Magnitude Bode Plot

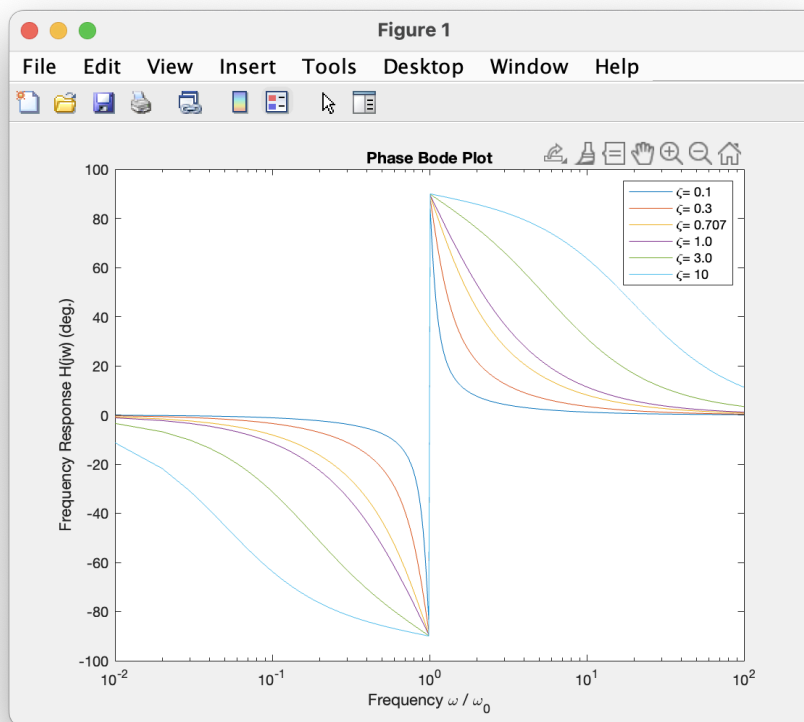


Figure 9: MF Phase Bode Plot

By looking at the magnitude, we can assume the circuit can be used to attenuate a specific frequency band, seen in the middle, while allowing frequencies outside that band to pass.

Problem 4: Linear System Tools: MATLAB

For the last part, useful commands were provided in the PDF that allowed us to analyze linear systems. These commands included creating a step response graph and creating bode plots (which are different from the log functions we used).

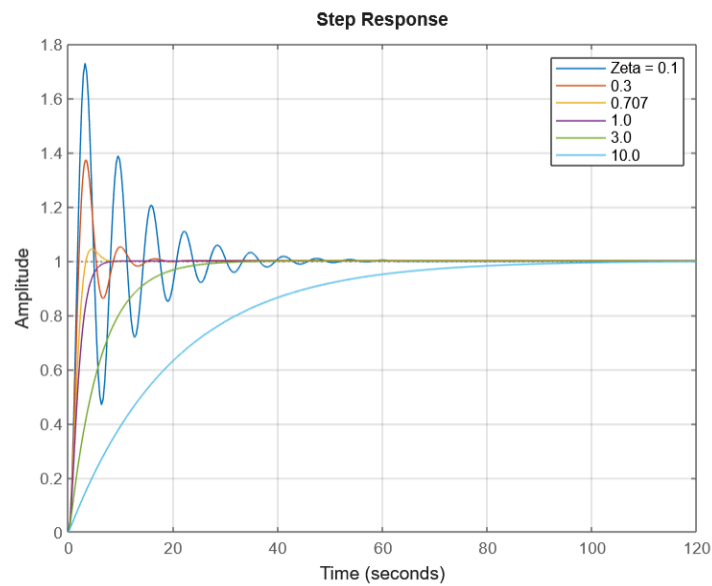


Figure 10: Step-Response Transfer Functions

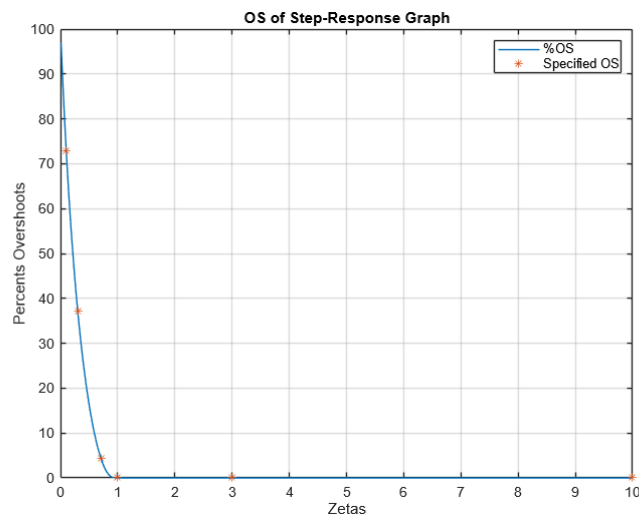


Figure 11: Overshoot of Step-Response Graph

Using these new functions, we looked back to part 1c and saw new ways of looking at the LPF transfer function. We calculated what are called "overshoot values," the phenomenon where the output of a system exceeds the desired or steady-state value during a transient response.

Conclusion

In conclusion, we saw how useful MATLAB was in realizing the properties of differing filters. As well as this, we saw the effect that the general equations involving zeta have useful information for us to understand on Bode plots, thus allowing us to fully comprehend the graphs.

MATLAB Code

```
Editor - /Users/diegoceja/Documents/MATLAB/Lab1.m
Lab1.m
1 w0 = 1;
2 zeta = [0.1 0.3 0.707 1.0 3.0 10.0];
3 w = 0.01: 0.01: 100;
4
5 for i = 1:length(zeta) %measuring the length of zeta
6     zeta_value = zeta(i); %Getting zeta point
7
8     h = 1 ./ (1 + 2*zeta_value*((1i*w)/w0) + ((1i*w)/w0).^2); %Getting freq. response
9
10    h2 = abs(h); %getting the mag of freq. response
11
12    plot(w,h2); %plotting w vs mag of freq. response
13
14    xlabel("Frequency \omega / \omega_0");
15    ylabel("Frequency Response |H(jw)|");
16    title("Linear |H(jw)| Plot");
17    legend(["\zeta= 0.1", "\zeta= 0.3", "\zeta= 0.707", "\zeta= 1.0", "\zeta= 3.0", "\zeta= 10"]);
18    hold on;
19 end
20
21
```

LPF Linear Code

```
Editor - /Users/diegoceja/Documents/MATLAB/Lab1.m
Lab1.m
1 w0 = 1;
2 zeta = [0.1 0.3 0.707 1.0 3.0 10.0];
3 w = 0.01: 0.01: 100;
4
5 for i = 1:length(zeta) %measuring the length of zeta
6     zeta_value = zeta(i); %Getting zeta point
7
8     h = 1 ./ (1 + 2*zeta_value*((1i*w)/w0) + ((1i*w)/w0).^2); %Getting freq. response
9
10    ph = rad2deg(angle(h)); %% getting the phase of the freq. response
11
12    semilogx(w,ph); %plotting w vs phase of freq. response
13
14    xlabel("Frequency \omega / \omega_0");
15    ylabel("Frequency Response \angle H(jw) (deg.)");
16    title("Phase Bode Plot");
17    legend(["\zeta= 0.1", "\zeta= 0.3", "\zeta= 0.707", "\zeta= 1.0", "\zeta= 3.0", "\zeta= 10"]);
18    hold on;
19 end
20
21
```

LPF Phase Code

```
Editor - /Users/diegoceja/Documents/MATLAB/Lab1.m
Lab1.m
1 w0 = 1;
2 zeta = [0.1 0.3 0.707 1.0 3.0 10.0];
3 w = 0.01: 0.01: 100;
4
5 for i = 1:length(zeta) %measuring the length of zeta
6     zeta_value = zeta(i); %Getting zeta point
7
8     h = 1 ./ (1 + 2*zeta_value*((1i*w)/w0) + ((1i*w)/w0).^2); %Getting freq. response
9
10    h2 = abs(h); %getting the mag of freq. response
11
12    hdb = 10*log10(h2.^2); %% changing the mag to dB
13
14    semilogx(w,hdb); %plotting w vs mag of freq. response
15
16    xlabel("Frequency \omega / \omega_0");
17    ylabel("Frequency Response |H(jw)| (dB)");
18    title("Magnitude Bode Plot");
19    legend(["\zeta= 0.1", "\zeta= 0.3", "\zeta= 0.707", "\zeta= 1.0", "\zeta= 3.0", "\zeta= 10"]);
20    hold on;
21 end
22
23
```

LPF Mag. Code

```

Editor - /Users/diegoceja/Documents/MATLAB/Lab1.m
Lab1.m x Lab1phase.m x +
1 w0 = 1;
2 zeta = [0.1 0.3 0.707 1.0 3.0 10.0];
3 w = 0.01: 0.01: 100;
4
5 for i = 1:length(zeta) %measuring the length of zeta
6     zeta_value = zeta(i); %Getting zeta point
7
8     h = (((1i*w)/w0).^2) ./ (1 + 2*zeta_value*((1i*w)/w0) + ((1i*w)/w0).^2); %Getting freq. response
9
10    h2 = abs(h); % getting the phase of the freq. response
11
12    hdb = 10*log10(h2.^2);
13
14    semilogx(w,hdb); %plotting w vs phase of freq. response
15
16    xlabel("Frequency \omega / \omega_0");
17    ylabel("Frequency Response (dB)");
18    title("Magnitude Bode Plot");
19    legend(["\zeta= 0.1", "\zeta= 0.3", "\zeta= 0.707", "\zeta= 1.0", "\zeta= 3.0", "\zeta= 10"]);
20    hold on;
21 end
22

```

HPF Mag. Code

```

Editor - /Users/diegoceja/Documents/MATLAB/Lab1phase.m
Lab1.m x Lab1phase.m x +
1 w0 = 1;
2 zeta = [0.1 0.3 0.707 1.0 3.0 10.0];
3 w = 0.01: 0.01: 100;
4
5 for i = 1:length(zeta) %measuring the length of zeta
6     zeta_value = zeta(i); %Getting zeta point
7
8     h = (((1i*w)/w0).^2) ./ (1 + 2*zeta_value*((1i*w)/w0) + ((1i*w)/w0).^2); %Getting freq. response
9
10    ph = rad2deg(angle(h)); % getting the phase of the freq. response
11
12    semilogx(w,ph); %plotting w vs phase of freq. response
13
14    xlabel("Frequency \omega / \omega_0");
15    ylabel("Frequency Response H(jw) (deg.)");
16    title("Phase Bode Plot");
17    legend(["\zeta= 0.1", "\zeta= 0.3", "\zeta= 0.707", "\zeta= 1.0", "\zeta= 3.0", "\zeta= 10"]);
18    hold on;
19 end
20
21

```

HPF Phase Code

```

Editor - /Users/diegoceja/Documents/MATLAB/Lab1.m
Lab1.m x Lab1phase.m x +
1 w0 = 1;
2 zeta = [0.1 0.3 0.707 1.0 3.0 10.0];
3 w = 0.01: 0.01: 100;
4
5 for i = 1:length(zeta) %measuring the length of zeta
6     zeta_value = zeta(i); %Getting zeta point
7
8     h = (((1i*w)/w0).^2 + 1) ./ (1 + 2*zeta_value*((1i*w)/w0) + ((1i*w)/w0).^2); %Getting freq. response
9
10    h2 = abs(h); % getting the phase of the freq. response
11
12    hdb = 10*log10(h2.^2);
13
14    semilogx(w,hdb); %plotting w vs phase of freq. response
15
16    xlabel("Frequency \omega / \omega_0");
17    ylabel("Frequency Response (dB)");
18    title("Magnitude Bode Plot");
19    legend(["\zeta= 0.1", "\zeta= 0.3", "\zeta= 0.707", "\zeta= 1.0", "\zeta= 3.0", "\zeta= 10"]);
20    hold on;
21 end
22
23

```

MF Mag. Code


```

Editor - /Users/diegoceja/Documents/MATLAB/Lab1phase.m
Lab1.m x Lab1phase.m x +
1 w0 = 1;
2 zeta = [0.1 0.3 0.707 1.0 3.0 10.0];
3 w = 0.01: 0.01: 100;
4
5 for i = 1:length(zeta) %measuring the length of zeta
6     zeta_value = zeta(i); %Getting zeta point
7
8     h = (((1i*w)/w0).^2 + 1) ./ (1 + 2*zeta_value*((1i*w)/w0) + ((1i*w)/w0).^2); %Getting freq. response
9
10    ph = rad2deg(angle(h)); % getting the phase of the freq. response
11
12    semilogx(w,ph); %plotting w vs phase of freq. response
13
14    xlabel("Frequency \omega / \omega_0");
15    ylabel("Frequency Response H(jw) (deg.)");
16    title("Phase Bode Plot");
17    legend(["\zeta= 0.1", "\zeta= 0.3", "\zeta= 0.707", "\zeta= 1.0", "\zeta= 3.0", "\zeta= 10"]);
18    hold on;
19 end
20
21

```

MF Phase Code

```

Editor - /Users/diegoceja/Documents/MATLAB/Lab1.m
Lab1.m x untitled * x +
1 w0 = 1;
2 zeta = [0.1 0.3 0.707 1.0 3.0 10.0];
3 w = 0.01: 0.01: 100;
4
5 for i = 1:length(zeta) %measuring the length of zeta
6     zeta_value = zeta(i); %Getting zeta point
7
8     h = (2*zeta_value*((1i*w)/w0)) ./ (1 + 2*zeta_value*((1i*w)/w0) + ((1i*w)/w0).^2); %Getting freq. response
9
10    ph = rad2deg(angle(h)); % getting the phase of the freq. response
11
12    semilogx(w,ph); %plotting w vs phase of freq. response
13
14    xlabel("Frequency \omega / \omega_0");
15    ylabel("Frequency Response \angle H(jw) (deg.)");
16    title("Phase Bode Plot");
17    legend(["\zeta= 0.1", "\zeta= 0.3", "\zeta= 0.707", "\zeta= 1.0", "\zeta= 3.0", "\zeta= 10"]);
18    hold on;
19 end
20
21

```

BPF Phase Code

```

Editor - /Users/diegoceja/Documents/MATLAB/Lab1.m
Lab1.m x untitled * x +
1 w0 = 1;
2 zeta = [0.1 0.3 0.707 1.0 3.0 10.0];
3 w = 0.01: 0.01: 100;
4
5 for i = 1:length(zeta) %measuring the length of zeta
6     zeta_value = zeta(i); %Getting zeta point
7
8     h = (2*zeta_value*((1i*w)/w0)) ./ (1 + 2*zeta_value*((1i*w)/w0) + ((1i*w)/w0).^2); %Getting freq. response
9
10    h2 = abs(h); %% getting the phase of the freq. response
11
12    hdb = 10*log10(h2.^2);
13
14    semilogx(w,hdb); %plotting w vs phase of freq. response
15
16    xlabel("Frequency \omega / \omega_0");
17    ylabel("Frequency Response (dB)");
18    title("Magnitude Bode Plot");
19    legend(["\zeta= 0.1", "\zeta= 0.3", "\zeta= 0.707", "\zeta= 1.0", "\zeta= 3.0", "\zeta= 10"]);
20    hold on;
21 end
22
23

```

BPF Mag. Code

```

% Problem 4a

w0 = 1; % fundamental frequency
w = 0.01:0.01:100; % store samples of omega
zeta = [0.1 0.3 0.707 1.0 3.0 10.0]; % dampening ratios

s=tf('s'); % s is the transfer function variable

for i=1:length(zeta)
    h = 1/(1 + 2*zeta(i)*s/w0 + (s/w0)^2); % H(jw)

    figure(1);
    step(h);
    hold on;
    grid on;
    legend('Zeta = 0.1','0.3','0.707','1.0','3.0','10.0');
end

```

Step-Response Transfer Functions

Code

```

% Problem 4b
clc
w0 = 1; % fundamental frequency
w = 0.01:0.01:100; % store samples of omega
zeta = [0.1 0.3 0.707 1.0 3.0 10.0]; % dampening ratios
zetas = 0.01:0.01:10; % from 0.01 to 10, as required from the PDF
s=tf('s'); % s is the transfer function variable

per_OS = zeros(size(zetas)); % Initialize overshoots
specificOS = zeros(size(zeta));

% If greater than 1, = 0
per_OS = 100.*exp(-pi.*zetas./sqrt(1-zetas.^2)).*(zetas<1);

for i=1:length(zeta)
    h(i,:) = 1/(1 + 2*zeta(i)*s/w0 + (s/w0)^2); % H(jw)
end
specificOS = [stepinfo(h).Overshoot] % Return only Overshoot
plot(zetas,per_OS);
hold on;

plot(zeta, specificOS, '*')

xlabel('Zetas');
ylabel('Percents Overshoots');
title('OS of Step-Response Graph');
legend('%OS','Specified OS');
grid on;
hold off;

```

Overshoot of Step-Response Graph

Code